

Literature Benchmark Brief: Nielsen_2006_TPWRS

Title: Using quantile regression to estimate the wind power forecast distribution

Authors: H. A. Nielsen, H. Madsen, T. S. Nielsen (2006)

Venue: IEEE Transactions on Power Systems | **DOI:** 10.1109/TPWRS.2006.873111

Cluster: Cluster D: Probabilistic Sizing & Reserve

Core Theoretical Mechanism & Model Formulation

Spline-based quantile regression for wind power forecast distribution estimation

Empirical Scale & Experimental Dataset

Western Denmark transmission system (Eltra)

Relevance to IEEE TSTE & Sustainable Energy Systems

Demonstrated that wind forecast distribution is skewed and variance varies with wind speed

Identified Gaps & Operational Vulnerabilities

Parametric spline smoothing blurs sharp pitch control transition cliffs

Role in Manuscript Defense & Reviewer #2 Proof

Theoretical justification for state-conditional residual quantile modeling